Chunking Verdicts in Retrieval-Augmented Generation Depend on the Evaluation Protocol: A Multiverse Analysis

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Abstract

Studies that compare chunking strategies for retrieval-augmented generation often retrieve a fixed number of chunks, so longer chunks also bring more text. Under evidence-coverage metrics, that extra text can look like better segmentation. We measure how much this matters with a multiverse analysis over four English question-answering datasets (9,253 held-out queries), scoring word-level evidence spans. When retrieval is matched on a word budget instead of a chunk count, the share of settings in which 512-word chunks significantly outperform 128-word chunks falls by 70.9 percentage points (95% CI [64.8, 81.9]) within the evaluated grid. For semantic-breakpoint chunking, the share of settings favoring the shorter arm rises by 55.6 points (95% CI [50.8, 63.1]). Both results hold when documents rather than queries are the sampling unit. Clustering-based chunking and dense retrieval are also sensitive to the protocol, but our original dense-retrieval hypothesis is not established: its estimate lies at the decision threshold, and its verdict depends on the inference unit. The advantage of paragraph-aware chunking is concentrated in queries whose evidence fixed windows split, which suggests, without establishing it, that part of the advantage reflects chunk boundaries coinciding with the datasets' annotation units. An exploratory answer-generation check with a Qwen3-4B reader shows positive coverage interactions but inconclusive answer-F1 interactions, and for the primary comparison exact match moves in the opposite direction. A chunking preference is therefore tied to the retrieval constraint under which it was measured; we recommend reporting both retrieval depth and text budget, and measuring answer quality directly.

1 Introduction

Retrieval-augmented generation (RAG) combines a language model with material retrieved from a document collection [9, 14]. Chunking determines which parts of that collection can be ranked and passed to the model. A useful comparison of chunkers must therefore answer two questions: what changes between the systems, and what remains fixed?

Consider two systems that each retrieve five passages. One uses a cap of 128 words per chunk; the other uses 512. Equal retrieval depth gives the second system a much larger potential text allowance. A metric that asks whether retrieved text contains a gold evidence span can reward that allowance, even when the longer chunks are not ranked more precisely. Matching the retrieved word budget removes this difference, but also changes the number of passages and can truncate the last selected chunk. The two protocols evaluate different operating conditions.

We ask how strongly this choice affects the direction and statistical support of a chunking comparison. Rather than choose one dataset, retriever and retrieval depth, we evaluate a declared grid of settings through a multiverse analysis [25]. Each setting, or *universe*, produces a paired comparison between two chunkers. We summarize the share of universes supporting each arm under count-matched and word-budget-matched selection.

The primary comparison is between `fixed-128` and `fixed-512`. Support for the larger arm falls by 70.9 pp under budget matching. The overlap comparisons change much less, while paragraph and semantic chunkers show substantial sensitivity when their realized lengths differ from those of

the fixed windows. A dense-retrieval extension shows that the starting preference also depends on the retriever. Finally, an answer-generation experiment tests whether the retrieval pattern carries over to the end task.

Our contribution is an estimate of protocol sensitivity across two studies with written analysis plans, post-hoc checks of what drives it, and an answer-level check that probes the limits of the retrieval result. We do not propose a new chunking algorithm, and we do not treat one protocol as universally correct. The aim is to make explicit the constraint under which a chunking preference holds.

2 Related work

Chunking strategies and their evaluation. Chunk-size comparisons, boundary-selection methods and context-aware embeddings address different parts of the retrieval problem. Bhat et al. [3] study fixed-size chunks in long-document retrieval. Financial-report chunking [12] and LumberChunker [8] investigate alternatives to uniform segmentation. Late chunking instead changes when chunk representations are formed relative to contextual encoding [10]. These distinctions matter when interpreting a comparison: an improvement in chunk representation is not necessarily an improvement in boundary selection.

Qu et al. [19] provide a close reference for our semantic-chunking extension. Their Section 2 defines fixed chunks by grouping consecutive sentences and compares them with breakpoint and clustering methods. Their retrieval evaluation uses F1, balancing precision and recall after chunks are mapped to documents or evidence sentences (their Section 4.1 and Appendix D). Their Appendix E.3 discusses how chunk size affects those components. Our study instead uses word caps and coverage-oriented span metrics. The breakpoint setting and capped clustering procedure are adaptations of the chunker families, not a replication of that paper's experimental configuration. We therefore test protocol sensitivity under our design rather than claim to overturn its conclusions.

Controlling text volume. Length control is not new. MoC holds average chunk length consistent across methods in its experimental settings [28]. HiChunk uses a retrieved-context token budget rather than simply equating the number of chunks [15]. Chen et al. compare passage, sentence and proposition units for dense retrieval and report that finer units improve retrieval-augmented question answering under a fixed computation budget [5]. These controls address different quantities: average chunk length describes a corpus, whereas a retrieval budget limits the text selected for a query. We ask how comparison outcomes change when only the selection protocol varies over a common evaluation grid.

Evaluation choices and uncertainty. BERGEN organizes RAG evaluation around explicit retrieval, generation and scoring choices [21]. Work on open-domain answer evaluation shows why a single answer metric can be an incomplete measure of correctness [13]. We keep retrieval and answer effects separate for the same reason.

Multiverse analysis and specification-curve analysis expose a result's dependence on analytical choices [23, 25]; related work studies this dependence in machine learning [1]. Dodge et al. show that reporting expected validation performance as a function of the computation spent on model development (hyperparameter-search trials or training time) can change the conclusion of a model comparison [7]. Research on significance testing and uncertainty in information retrieval and language-model evaluation motivates our paired design and explicit power assumptions [17, 24, 26].

These methods quantify uncertainty within a design. They do not make the chosen datasets or settings representative of every deployment.

3 Problem formulation

3.1 Chunks, selected text and budgets

We define a word as a maximal non-whitespace run. A chunk is represented by half-open word intervals and the identity of its document. Most chunkers in this study return one contiguous interval per chunk; clustering can join non-adjacent intervals. Let $w(c)$ be the number of words in chunk c, and let $r_{q,C}(i)$ be the chunk ranked at position i for query q under chunker C.

Count-matched selection returns the first k chunks. Counting every selected word occurrence, its text allowance is

$$T_{q,C}^{(k)} = \sum_{i=1}^{k} w(r_{q,C}(i)). \tag{1}$$

For complete fixed-length chunks, this is kW. Terminal fragments and other packing rules break this equality. Corpus-average chunk length is therefore a useful diagnostic, but it does not determine the length of every query's top-ranked chunks.

Budget-matched selection takes chunks in rank order until B words have been selected, truncating the final chunk to the remaining allowance. Overlapping text is charged each time it appears. Its allowance $T_{q,C}^{(B)}$, the number of selected word occurrences, equals B when enough text is available. Intervals within a non-contiguous chunk are traversed in document order before truncation.

Let $R_{q,C}$ be the set of document-tagged word positions selected under the protocol in use. Evidence is scored against this set, which counts a repeated position only once. Thus $R_{q,C}$ and the allowance ($T_{q,C}^{(k)}$ or $T_{q,C}^{(B)}$) measure different things: unique word positions and word occurrences, respectively. Equal word budgets need not produce equal unique-text coverage.

3.2 Span metrics

For the nonempty set of gold evidence spans G_q, we compute

$$\mathrm{hit}_{\mathrm{strict}}(q, C) = \mathbb{1}\{\exists g \in G_q : g \subseteq R_{q,C}\}, \tag{2}$$

$$\mathrm{hit}_{\mathrm{lenient}}(q, C) = \mathbb{1}\{\exists g \in G_q : |g \cap R_{q,C}| \geq 0.5|g|\}, \tag{3}$$

$$\mathrm{coverage}(q, C) = \max_{g \in G_q} \frac{|g \cap R_{q,C}|}{|g|}. \tag{4}$$

Lengths here are measured in words. Gold evidence is annotated as character spans; each span is mapped to the half-open interval of words it overlaps, so a span boundary inside a word extends the span to that whole word, and a span that touches only whitespace is dropped. The analysis plans define the three metrics over characters; the stored results, and every result reported here, use this word mapping. Appendix A lists this as a deviation. A separate post-hoc analysis re-executed BM25 retrieval, confirmed that it reproduces every stored score and word count, and rescored those selections with character-based definitions; it changed no hypothesis verdict (Appendix D). Extending the set of selected positions cannot reduce any of these scores. This monotonicity does not imply that longer chunks from a different ranking must retrieve more evidence; it explains why an increased text allowance can help these metrics without improving precision.

The metrics reward any qualifying span, or the best-covered span. They neither require all spans needed for a multi-part answer nor penalize irrelevant text. Their scope is evidence acquisition under these definitions, not the reader's ability to use the evidence.

3.3 Verdicts and support shares

A universe u specifies a dataset, retriever, protocol, operating point and metric. For arms A and B, the paired query-level difference is

$$\Delta_{u,q} = m_u(A, q) - m_u(B, q). \tag{5}$$

The verdict supports A when the 95% paired interval for the mean difference lies above zero, supports B when it lies below zero, and is inconclusive otherwise. An inconclusive interval is not evidence of equivalence. The interval is $\bar{\Delta}_u \pm c\,\widehat{\mathrm{se}}(\bar{\Delta}_u)$ with the standard error computed at the sampling unit. With queries as the unit, it is the standard deviation of $\Delta_{u,q}$ divided by $\sqrt{n}$, and $c = 1.96$. With documents as the unit, it is the cluster-robust (CR1) standard error over the G documents, and c is the 97.5th percentile of the t distribution with $G - 1$ degrees of freedom.

For the designed set of universes $\mathcal{U}_P$ under protocol P, define

$$s_P(A) = \frac{1}{|\mathcal{U}_P|} \sum_{u \in \mathcal{U}_P} \mathbb{1}\{u \text{ supports } A\}. \tag{6}$$

The figures show the net share $s_P(A) - s_P(B)$. The primary H1 statistic is the decline in larger-arm support:

$$D_{\mathrm{H1}} = 100\{s_{\mathrm{count}}(\texttt{fixed-512}) - s_{\mathrm{budget}}(\texttt{fixed-512})\} \text{ pp}. \tag{7}$$

The factor 100 converts fractions to percentage points. Count and budget grids have different operating points and may have different sizes. A difference in their support shares is therefore not a paired verdict-flip rate, a percentage improvement in retrieval score, or an answer-accuracy gain.

4 Study 1: chunk size and structure-aware chunking

4.1 Data and design

We use four English question-answering collections with located answer or evidence spans: SQuAD 2.0 [20], COVID-QA [18], CUAD [11] and Qasper [6]. Table 1 gives the corpus and evaluation counts.

For SQuAD, we retain answerable development questions and join the paragraphs of each article into a document. CUAD uses answerable clauses from the version 1 test set, allowing multiple spans. Qasper combines development and test questions with at least one evidence paragraph located in the document; table- and figure-only evidence is excluded. Evidence offsets refer to the assembled document strings used by the experiment.

Documents are assigned to exploratory and confirmatory splits in an approximately 20/80 ratio, using seed 20261018. Queries follow their document. The full collection remains indexed for retrieval: the split separates exploratory from confirmatory evaluation, not the documents that can be retrieved. Both retrieval studies use the same 9,253 confirmatory queries.

Table 1: Datasets. Documents are split 20/80 into exploratory and confirmatory sets (seed 20261018); queries follow their document. All results are on the confirmatory queries; the retrieval corpus is always the full dataset.

Dataset	Domain	Documents	Confirmatory docs	Confirmatory queries
SQuAD 2.0	Wikipedia	35	28	4,744
COVID-QA	biomedical papers	147	118	1,771
CUAD	legal contracts	102	82	996
Qasper	NLP papers	697	558	1,742
Total		981	786	9,253

Chunkers. We evaluate 16 configurations: four caps $W \in \{64, 128, 256, 512\}$ applied to fixed windows, overlapping fixed windows, sentence packing and paragraph packing. Fixed windows use stride W without overlap and $0.75W$ with 25% overlap. Sentence packing adds whole sentences up to the cap, splitting an overlong sentence into fixed windows. Paragraph packing uses blank-line boundaries, with sentence packing as the fallback for an overlong paragraph. All Study 1 chunks are contiguous. Their mean lengths can be substantially below their caps (Table 8).

Retrievers and operating points. The lexical grid has 16 BM25 configurations [22]: Lucene-style BM25 or BM25+, $k_1 \in \{0.9, 1.5\}$, $b \in \{0.4, 0.75\}$, and either English stop-word removal with Snowball stemming or lowercase-only tokenization. We use `bm25s` 0.2.13 [16]. The default retriever is Lucene-style BM25 with $k_1 = 1.5$, $b = 0.75$ and stemming.

Count matching uses $k \in \{1, 3, 5, 10\}$. Budget matching uses $B \in \{128, 256, 512, 1024, 2048\}$ words. The main analysis admits only $B \geq \max(W_A, W_B)$, so neither arm must truncate its highest-ranked chunk. Later chunks can still be truncated. We retain the smaller budgets for a sensitivity analysis. Per-query metric values are stored in half precision (float16); re-executing the BM25 retrieval with full-precision storage changes no universe verdict (Appendix D).

P1 compares `fixed-128` with `fixed-512`. S1 compares neighboring fixed sizes W and $2W$. S2 compares overlap with no overlap at the same cap; S3 compares paragraph packing with fixed windows. S3-256 is designated for H1b and H2. S4, sentence packing versus fixed windows, is exploratory. For P1, the factorial design yields $4 \times 16 \times 4 \times 3 = 768$ count-matched and $4 \times 16 \times 3 \times 3 = 576$ admissible budget-matched universes.

4.2 Analysis plan

Study 1 follows a written analysis plan that fixes its hypotheses, thresholds and analyses; the plan was not deposited with an external registry. The pilot used only exploratory documents. Appendix A lists the changes recorded in the plan's deviation log, including a correction to one sensitivity-analysis interval, the places where the implementation differs from the plan's text, and the correction of the inference procedure described below.

H1, the single primary hypothesis of Study 1, predicts a decline of at least 20 percentage points in support for `fixed-512` under budget matching; H1b, H2 and H3 are secondary. The plan's support rule requires a point estimate of at least 20 percentage points and a 95% interval excluding zero; an upper bound below 20 falsifies the threshold claim. This rule does not require the lower bound itself to exceed 20. H1b applies the corresponding 10-point threshold to the increase in support for `paragraph-256`.

H2 predicts that dataset has the largest main-effect η^2 for S3-256 under budget matching in at least 95% of bootstrap replicates. The decomposition is a main-effects analysis of variance of the

universe mean differences on dataset, retriever, budget and metric.

H3 concerns the approximate paired sample size needed to detect a difference of $|\delta| = 0.02$ at two-sided $\alpha = 0.05$ and power 0.8. With paired-difference standard deviation σ_d, we use

$$n = \left(\frac{(1.96 + 0.8416)\sigma_d}{0.02} \right)^2 . \tag{8}$$

H3 predicts $n > 500$ in more than half of the budget-matched S2/S3 universes. This is a variance-based planning calculation, not an experiment in which additional queries were collected.

The same verdict rule (Section 3.3) defines every point estimate and every bootstrap replicate. Support shares are correlated because universes reuse queries, so their uncertainty comes from 1,000 joint replicates (seed 11): in each replicate, the sampling units are resampled within each dataset, the same draw is applied to all of that dataset's universes, and every universe's verdict is recomputed with the rule. Evaluated on the original sample, the replicate statistic is the point estimate. H2's replicates use the same scheme (seed 13). The 95% interval is the 2.5th to 97.5th percentile range of the replicate statistic. Because a share changes in discrete steps when verdicts cross the decision boundary, these intervals are approximate: they need not be centered on the estimate (replicate means differ from the estimates by up to 6.3 percentage points), and an upper bound can exceed the largest decline the observed count-matched share allows. We report every hypothesis with queries as the resampling and variance unit, the unit fixed by the plans, and again with documents, which respects the nesting of queries in documents. Both versions are conditional on the four datasets and the designed grid. Universes have equal weight; in this balanced grid, datasets consequently have equal representation despite different query counts.

The plans specified a different computation: point-estimate verdicts from 95% percentile-bootstrap intervals (2,000 query resamples, seed 7), and normal-approximation verdicts inside the joint replicates, which used the query-level standard deviation even when documents were resampled. The point estimate and its interval then describe slightly different statistics, and the document-level interval ignores within-document dependence in its verdicts. We identified this after the results were known and report the consistent procedure as a correction. Table 7 gives every share hypothesis under both computations; estimates differ by at most 2.1 percentage points, and only H5's verdict differs (Section 5).

The plan specifies five sensitivity analyses: (a) Benjamini–Hochberg adjustment at 5% [2], (b) excluding SQuAD, (c) the default retriever only, (d) resampling documents instead of queries, and (e) admitting all budgets. The adjustment uses the two-sided p-values of the Wald rule and is applied separately to the universes of each claim, protocol and dataset; the plan's wording says "within claim", and pooling the family over datasets would give 70.2 pp instead of 67.6 pp (post-hoc check). Sensitivity analysis (d) resamples documents and uses the document-level verdict rule, so it equals the document-unit analysis of H1. Benjamini–Hochberg controls the false-discovery rate under independence or positive dependence among tests; we use it as a robustness check, not as a guarantee for these correlated universes.

4.3 Results

The primary contrast. Under count matching, the 512-word arm receives support in 78.5% of universes and the 128-word arm in 0.0%. Under budget matching, the larger arm's share is 7.6%, while the smaller arm's is 81.2% (Table 2). The larger-arm decline is 70.9 pp (95% CI [64.8, 81.9]), so H1 is supported. Shares and their difference are rounded independently. Figure 1 shows the net shares for the main Study 1 contrasts; Table 2 lists all sixteen.

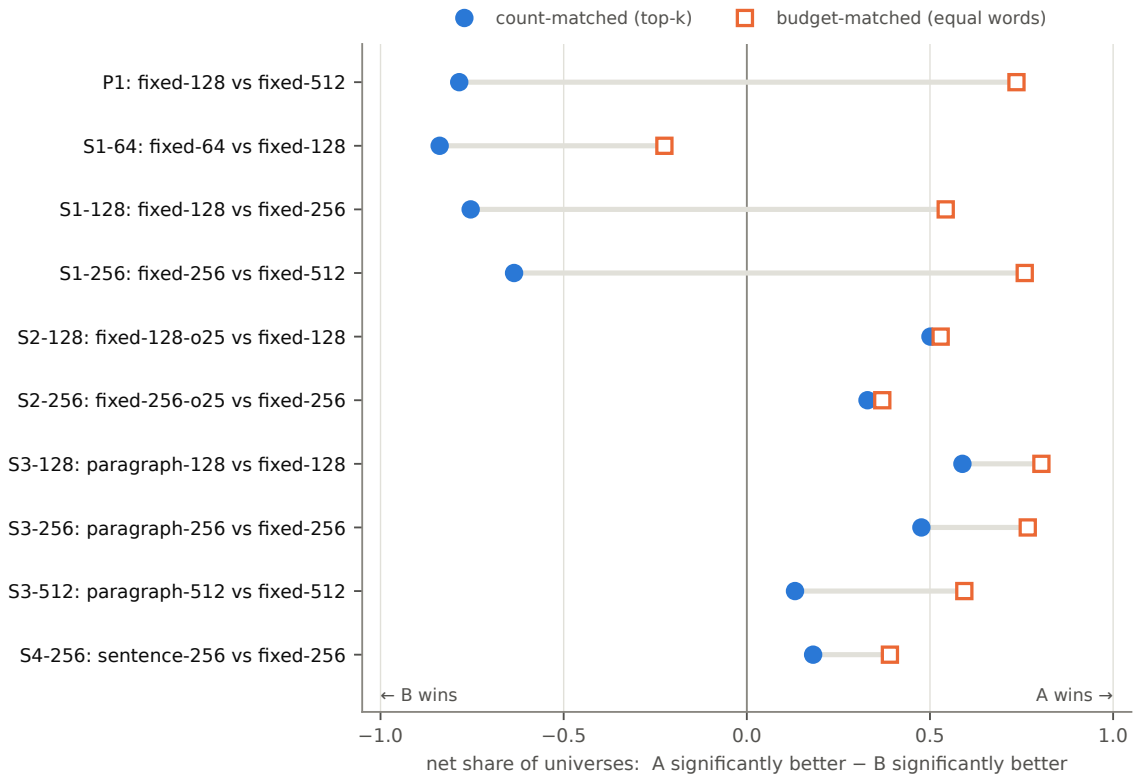


Figure 1: Study 1 net support shares, $s_P(A) - s_P(B)$, for selected contrasts on the confirmatory split. Circles show count matching; open squares show word-budget matching. Movement to the right favors A relative to B, but need not reverse the favored arm.

Table 2: Study 1: share of universes (%) whose 95% paired interval supports arm A or arm B, per claim and selection protocol (confirmatory split); the remainder is inconclusive. Budget-matched universes use the admissible budgets $B \geq$ the larger chunk size. n = number of universes.

Claim	Role	A vs B	A (count)	B (count)	n	A (budget)	B (budget)	n
P1	primary	`fixed-128` vs `fixed-512`	0.0	78.5	768	81.2	7.6	576
S1-64	secondary	`fixed-64` vs `fixed-128`	0.0	83.9	768	20.0	42.5	960
S1-128	secondary	`fixed-128` vs `fixed-256`	0.0	75.4	768	60.8	6.5	768
S1-256	secondary	`fixed-256` vs `fixed-512`	0.0	63.5	768	78.0	2.1	576
S2-64	secondary	`fixed-64-o25` vs `fixed-64`	52.5	1.7	768	55.3	1.7	960
S2-128	secondary	`fixed-128-o25` vs `fixed-128`	52.0	1.8	768	54.8	1.9	960
S2-256	secondary	`fixed-256-o25` vs `fixed-256`	36.2	3.3	768	39.8	2.9	768
S2-512	secondary	`fixed-512-o25` vs `fixed-512`	9.6	8.3	768	12.0	7.1	576
S3-64	secondary	`paragraph-64` vs `fixed-64`	62.2	5.7	768	75.7	0.0	960
S3-128	secondary	`paragraph-128` vs `fixed-128`	60.0	1.2	768	80.4	0.0	960
S3-256	secondary	`paragraph-256` vs `fixed-256`	50.1	2.5	768	77.2	0.5	768
S3-512	secondary	`paragraph-512` vs `fixed-512`	14.7	1.6	768	60.9	1.6	576
S4-64	exploratory	`sentence-64` vs `fixed-64`	58.9	6.1	768	69.4	0.3	960
S4-128	exploratory	`sentence-128` vs `fixed-128`	49.9	0.9	768	58.2	0.0	960
S4-256	exploratory	`sentence-256` vs `fixed-256`	19.4	1.3	768	39.1	0.0	768
S4-512	exploratory	`sentence-512` vs `fixed-512`	6.4	0.5	768	21.4	0.3	576

The decline persists in all five sensitivity analyses: 67.6 pp with multiplicity adjustment, 61.2 pp without SQuAD, 70.8 pp with the default retriever, 68.9 pp with document resampling, and 73.9 pp with all budgets (Table 10). A post-hoc breakdown by metric gives declines of 69.3, 70.7 and 72.7 pp for strict containment, lenient hits and coverage, so H1 does not rest on one metric. The direction of the budget-matched comparison does depend on the metric when gold spans are long. Qasper's gold evidence consists of whole paragraphs (median shortest span 72.5 words), and there strict containment favors the 512-word arm in 91.7% of budget-matched universes and the 128-word arm in 0.0%, the reverse of the lenient and coverage verdicts: a 128-word window often cannot hold a whole paragraph.

The mean P1 difference changes sign between protocols in each dataset (Table 9). The strength of the change is not uniform. CUAD has relatively few decisive count-matched settings. Qasper has mixed budget-matched verdicts and only a small positive average difference. The coverage curves in Figure 2 show this variation rather than reducing it to a single pooled preference.

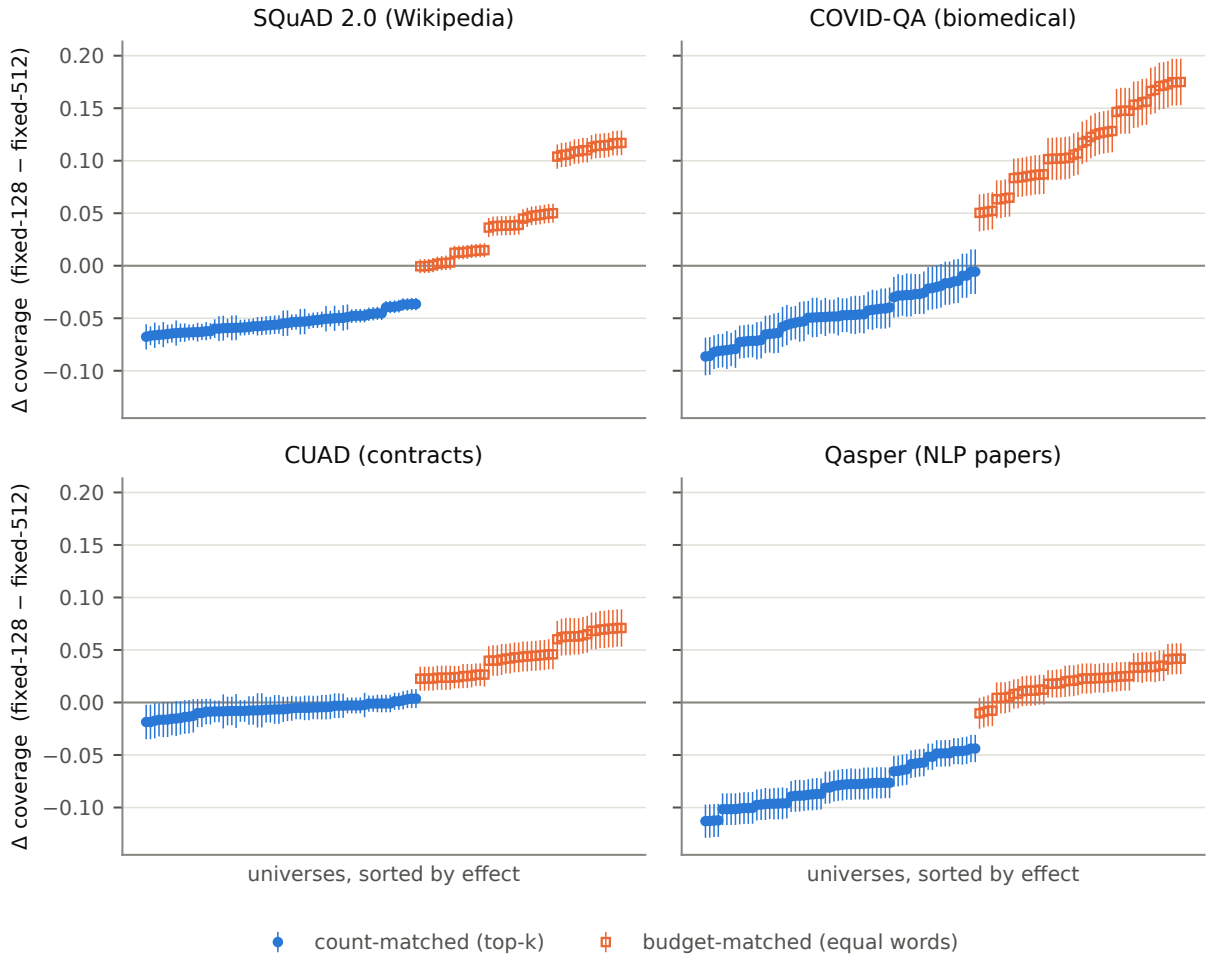


Figure 2: P1 coverage differences: `fixed-128` minus `fixed-512`, with 95% Wald intervals (query level). Universes are sorted by effect within each dataset and distinguished by selection protocol.

Overlap, structure and size. Overlap comparisons change much less. For S2-128, overlap support moves from 52.0% to 54.8%; for S2-256, from 36.2% to 39.8%. These arms share a

nominal cap, though terminal fragments produce slightly different mean lengths and overlap changes redundancy. Their relative stability is consistent with the text-allowance explanation but does not isolate it from other properties of the retrieved context.

For S3-256, paragraph-packing support rises by 27.1 pp (95% CI [15.4, 33.9]); H1b is supported. Paragraph packing produces shorter chunks on average than fixed windows at the same cap. Budget matching removes the unequal allowance in word occurrences, while leaving differences in boundaries and ranking intact. The effect is not uniform across domains (Figure 4).

Part of the paragraph advantage may come from how the datasets are built. Our documents join each dataset's own contexts or paragraphs with blank lines, and paragraph packing splits exactly there, so it rarely cuts a gold span. In a post-hoc count at a 256-word cap, at least one gold span lies inside a single chunk for 98.6% of Qasper queries under paragraph packing, against 75.5% for fixed windows (CUAD: 99.0% and 91.1%); sentence packing reaches 80.8% on Qasper. The advantage is concentrated where fixed windows split the evidence: under budget matching, the mean Qasper S3-256 difference over all budget-matched universes (all metrics) is 0.075 for the 24.5% of queries whose evidence `fixed-256` splits, and 0.015 for the others. An advantage remains where both chunkings contain the evidence (in SQuAD, where fixed windows split the evidence of only 23 queries, the mean difference among the others is 0.033), and the two groups of queries also differ in evidence length. Boundary alignment is therefore a plausible partial explanation, not an established cause, and the S3 contrasts should not be read as a general advantage of structure-aware chunking.

Smaller is not always better under budget matching. In S1-64, the 128-word arm retains 42.5% support, compared with 20.0% for the 64-word arm. The result is protocol dependence, not a rule to choose the smallest available chunk.

Dataset dependence and power. For S3-256 under budget matching, dataset has $\eta^2 = 0.61$, followed by budget and then the retriever ($\eta^2 = 0.04$), and is the largest factor in 100.0% of bootstrap replicates (100.0% with documents as the unit). H2 is supported; the more precise decomposition is in Table 11. Given the alignment effect above, a large dataset term is expected.

Across S2/S3 budget-matched universes, 85.4% have an estimated requirement above 500 paired queries, with median 1,256 (Table 12). H3 is supported. These estimates describe the observed paired-difference variance under the planning assumptions, not guaranteed power for a new document-clustered sample.

5 Study 1b: semantic chunking and dense retrieval

5.1 Design

Study 1b extends the configurations while retaining the datasets, confirmatory queries, metrics and analysis procedure. It is not an independent replication on new data. Its own written plan (version 1.0) specifies the hypotheses and analysis.

Semantic chunkers. The `semantic-bp` arm embeds sentences individually and inserts a boundary when the cosine distance between adjacent sentences exceeds the document's 95th percentile, calculated with linear interpolation. A one-sentence document becomes one chunk. Overlong sentences are split first; groups exceeding 512 words are repacked with the sentence rule. The percentile is fixed in our design rather than attributed to Qu et al.'s evaluated settings.

The `semantic-cl` arm uses the positional–semantic distance introduced by Qu et al. [19]:

$$d(i,j) = \lambda d_{\text{pos}}(i,j) + (1-\lambda) d_{\cos}(i,j), \tag{9}$$

$$d_{\text{pos}}(i,j) = |i-j|/n, \qquad d_{\cos}(i,j) = 1 - \max\{\cos(e_i, e_j), 0\}. \tag{10}$$

Here n is the number of sentences and e_i is a sentence embedding. We set $\lambda = 0.5$, visit pairs with $d \leq 0.5$ in increasing-distance order, break ties by sentence indices, and merge clusters only while their combined length remains at most 512 words. The resulting text can be non-contiguous; intervals are concatenated in document order. Qu et al. also stop merging at distance 0.5, but cap clusters by a number of sentences derived from the target number of chunks; the word cap makes this an adaptation of their procedure.

Both semantic chunkers embed sentences with BGE-M3, the dense retriever's encoder. We compare each with `fixed-128`, `fixed-256` and `fixed-512`, denoting breakpoint contrasts S5 and clustering contrasts S6. Breakpoint chunks average 278–320 words across datasets, while clustering chunks average 452–458. At least 92.0% of clustering chunks are non-contiguous (Table 8). This setting produces clusters near the cap, so the results need not generalize to other clustering configurations.

Retrieval and hypotheses. We use default BM25 and BGE-M3 dense retrieval [4]. Dense embeddings are normalized, stored in half precision and ranked by their dot product, which equals cosine similarity up to half-precision rounding. The maximum encoder input is 8,192 tokens. Word caps and token counts are different quantities, so the cap alone is not a check for encoder truncation.

For each dataset and contrast, the shorter arm is determined from corpus-average chunk lengths, a property of the chunking alone (Table 8). The breakpoint arm is longer than the 128- and 256-word fixed arms and shorter than the 512-word arm. Admissible budgets are 512, 1,024 and 2,048 words. Each S5/S6 contrast has 96 count-matched and 72 budget-matched universes; dense-only P1 has 48 and 36, respectively.

H4 predicts a shorter-arm support gain of at least 20 percentage points across breakpoint contrasts and retrievers. H4b uses a 10-point threshold for clustering. H5 retains H1's larger-arm formulation under dense retrieval: a decline of at least 20 percentage points in support for `fixed-512`. H4 is the single primary hypothesis of Study 1b; H4b, H5 and H5b are secondary.

In the exploratory pilot, the larger arm was already infrequently supported with dense retrieval (8.3% of count-matched universes), limiting how far its share could fall, while support for the shorter arm rose from 31.2% to 91.7%. The plan's deviation log records H5b as an amendment made in response: an increase of at least 20 percentage points in support for the shorter arm. H5b was therefore formulated after its effect had been observed on the exploratory split. Its confirmatory test uses different queries over the same indexed corpus, so it is weaker evidence than H4 and H4b. H5 remains the original test.

5.2 Results

Across breakpoint comparisons, shorter-arm support rises from 9.7% to 65.3%, a gain of 55.6 pp (95% CI [50.8, 63.1]). H4 is supported. The gain remains when retrievers are examined separately: 61.1 pp for BM25 and 50.0 pp for dense retrieval. Table 13 reports the sensitivity analyses, and Figure 3 shows the net shares of every Study 1b contrast.

These shorter-arm shares should not be read as semantic-arm shares. Against `fixed-128`, the breakpoint arm is supported in 60.4% of count-matched universes but only 8.3% of budget-matched universes. Against `fixed-512`, where the breakpoint arm is shorter, its support increases instead

(Table 3). Relative length changes the interpretation of the same semantic-versus-fixed comparison.

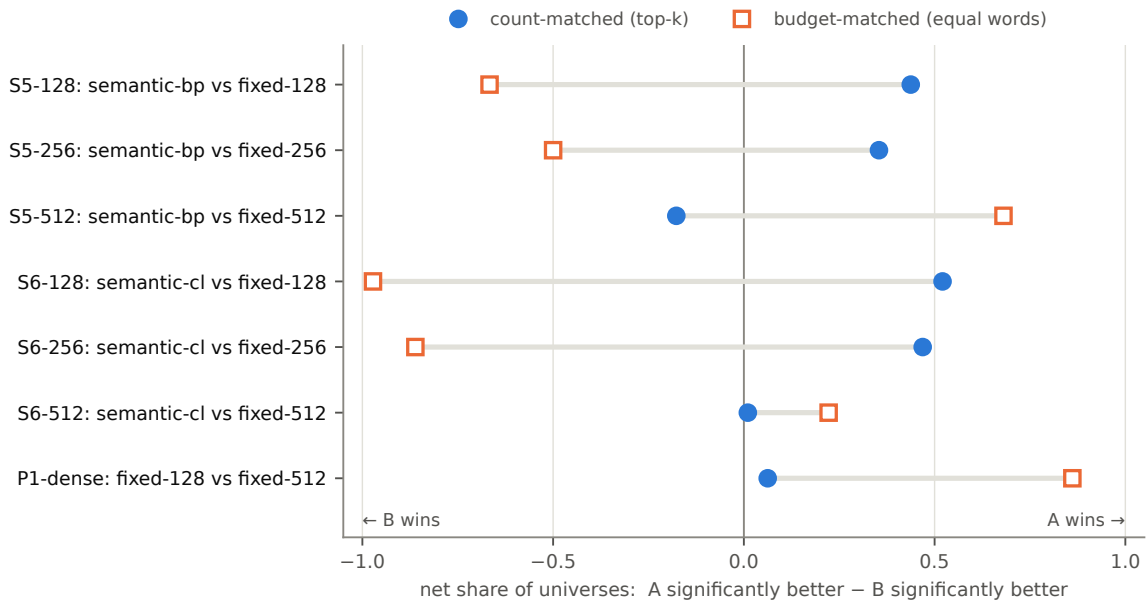


Figure 3: Study 1b net support shares. S5/S6 pool default BM25 and dense retrieval; P1-dense uses only BGE-M3. A and B remain fixed within each row, so movement must be read alongside the pair's relative lengths.

Table 3: Study 1b: share of universes (%) supporting arm A or B per claim and protocol (confirmatory). S5/S6 pool BM25 and dense retrieval; P1-dense uses the dense retriever only. Admissible budgets $B \in \{512, 1{,}024, 2{,}048\}$. The S5/S6 rows give semantic-arm (not shorter-arm) shares, the analogue of the semantic-versus-fixed comparisons in semantic-chunking studies (planned descriptive analysis g).

Claim	A vs B	A (count)	B (count)	n	A (budget)	B (budget)	n
S5-128	`semantic-bp` vs `fixed-128`	60.4	16.7	96	8.3	75.0	72
S5-256	`semantic-bp` vs `fixed-256`	39.6	4.2	96	2.8	52.8	72
S5-512	`semantic-bp` vs `fixed-512`	8.3	26.0	96	68.1	0.0	72
S6-128	`semantic-cl` vs `fixed-128`	54.2	2.1	96	0.0	97.2	72
S6-256	`semantic-cl` vs `fixed-256`	46.9	0.0	96	0.0	86.1	72
S6-512	`semantic-cl` vs `fixed-512`	18.8	17.7	96	33.3	11.1	72
P1-dense	`fixed-128` vs `fixed-512`	33.3	27.1	48	91.7	5.6	36

For clustering, shorter-arm support rises from 6.9% to 72.2%, a gain of 65.3 pp (95% CI [56.5, 68.3]). H4b is supported for this configuration. The comparison does not separately identify effects of length, clustering and non-contiguity.

For dense P1, larger-arm support falls from 27.1% to 5.6%. The decline of 21.5 pp (95% CI [16.7, 41.7]) meets the support rule, but only just. The dense P1 grid has 48 count-matched and 36 budget-matched universes, so one count-matched verdict moves the share by 2.1 percentage points, more than the margin above the threshold. With documents as the unit, the decline is 19.4 pp ([16.7, 39.6]) and H5 is inconclusive; under the plans' computation it was 19.4 pp and inconclusive. The difference between the computations comes from a single SQuAD universe whose percentile interval

ends exactly at zero. The interval runs from below the threshold to well above it, and its replicates average 5.5 points more than the estimate. We therefore treat H5 as unresolved, neither supported nor evidence against the hypothesis. The amended shorter-arm share rises from 33.3% to 91.7%, a gain of 58.3 pp ([45.1, 64.6]; with documents as the unit 58.3 pp, [47.9, 75.0]); H5b is supported at both units. The floor that motivated the amendment was less severe in the confirmatory data: the larger arm's count-matched support was 27.1%, against 8.3% in the pilot.

Table 4 summarizes the eight planned hypotheses. For each supported share hypothesis other than H5 (H1, H1b, H4, H4b and H5b), the lower bound of the interval also exceeds the threshold, at both units.

Table 4: Planned hypotheses and confirmatory outcomes, with queries (the planned unit) and with documents as the sampling unit. Share changes are percentage points, not retrieval-score improvements. H5b is a disclosed amendment formulated after the pilot. Table 7 adds the plans' original computation.

ID	Role	Quantity	Estimate [95% CI] or diagnostic	Verdict	Document unit
H1	primary	Larger-arm decline, P1	70.9 pp [64.8, 81.9]	supported	68.9 pp [61.6, 80.9], supported
H1b	secondary	Paragraph-256 gain	27.1 pp [15.4, 33.9]	supported	26.4 pp [15.0, 34.4], supported
H2	secondary	Dataset main effect	$\eta^2 = 0.61$, largest in 100.0%	supported	largest in 100.0%, supported
H3	secondary	Universes needing $n > 500$	85.4%, median $n = 1{,}256$	supported	not applicable
H4	primary	Shorter-arm gain, breakpoint	55.6 pp [50.8, 63.1]	supported	59.0 pp [51.3, 65.0], supported
H4b	secondary	Shorter-arm gain, clustering	65.3 pp [56.5, 68.3]	supported	65.2 pp [55.4, 67.1], supported
H5	secondary	Larger-arm decline, dense P1	21.5 pp [16.7, 41.7]	supported (marginal)	19.4 pp [16.7, 39.6], inconclusive
H5b	amendment	Shorter-arm gain, dense P1	58.3 pp [45.1, 64.6]	supported	58.3 pp [47.9, 75.0], supported

6 Exploratory answer-quality evaluation

6.1 Setup

We sampled 50 confirmatory queries per dataset using seed 20261019 and evaluated P1 and S3-256 with default BM25. The operating points were top-5 retrieval and a 1,024-word budget. A written generation analysis plan specifies the sample, the operating points and the analysis; the check is exploratory and does not change any confirmatory hypothesis verdict.

The reader is Qwen3-4B [27], using a Q4_K_M quantized GGUF model with greedy decoding, thinking disabled and a 256-token output limit. The prompt gives numbered passages and the question, requesting the shortest supported answer or exactly "unanswerable". We score answers by SQuAD-style token F1 and exact match against each question's gold answer strings: answer spans for SQuAD and COVID-QA, clause spans for CUAD, and Qasper's extractive, free-form or yes/no answers. Model and prompt details are in Appendix B.

For each query, let $\Delta_q(o)$ be the answer-score difference between A and B at operating point o. The difference-in-differences (DiD) is

$$\mathrm{DiD}_q = \Delta_q(B = 1{,}024) - \Delta_q(k = 5). \tag{11}$$

A positive DiD means that A gains relative to B when the protocol changes; it need not mean A has the higher score at either operating point. The pooled estimate weights datasets equally and uses 2,000 stratified bootstrap resamples. We compute the same contrast for retrieval coverage and report minimum detectable effects, $\mathrm{MDE} = (1.96 + 0.8416)\,\mathrm{SD}/\sqrt{n}$, with SD the standard deviation of the per-query DiD over the $n = 200$ queries.

6.2 Results

All scheduled generations completed. Coverage DiD is +0.089 (95% CI [+0.050, +0.132]) for P1 and +0.035 ([+0.015, +0.060]) for S3-256. Both contrasts are positive, consistent with the retrieval studies.

Answer F1 does not show a detectable positive interaction. The pooled DiDs are −0.017 ([−0.065, +0.033]) and +0.004 ([−0.024, +0.036]); both intervals include zero, so the sign of the answer-F1 interaction is unresolved, and the data do not establish that the protocol has no effect on answers. The intervals do exclude interactions as large as the minimum detectable effects (0.068 and 0.044) in either direction.

For P1, exact match moves in the opposite direction: −0.055 ([−0.100, −0.010]). The pooled cell means show the 128-word arm ahead at top-5 (0.105 versus 0.090), but the 512-word arm ahead at the matched budget (0.125 versus 0.085). These cell means are descriptive and have no separately reported intervals. Table 5 gives the full contrasts.

Table 5: Answer-quality check (exploratory; written analysis plan): difference-in-differences DiD = $\Delta(B{=}1{,}024) - \Delta(k{=}5)$, Δ = metric(A) − metric(B), with 95% bootstrap CIs (2,000 resamples; pooled: stratified by dataset). The retrieval mechanism predicts DiD > 0; an interval that includes 0 does not establish the absence of an effect. 50 queries per dataset; reader Qwen3-4B (Q4_K_M, greedy).

Claim	Dataset	Answer F1	Exact match	Coverage
P1	SQuAD 2.0	−0.122 [−0.245, +0.009]	−0.160 [−0.300, 0.000]	+0.120 [+0.040, +0.220]
	COVID-QA	−0.006 [−0.101, +0.090]	−0.060 [−0.160, +0.020]	+0.110 [+0.040, +0.200]
	CUAD	−0.001 [−0.031, +0.033]	0.000 [0.000, 0.000]	+0.020 [0.000, +0.060]
	Qasper	+0.060 [−0.015, +0.158]	0.000 [−0.060, +0.060]	+0.106 [+0.020, +0.214]
	Pooled (stratified)	−0.017 [−0.065, +0.033]	−0.055 [−0.100, −0.010]	+0.089 [+0.050, +0.132]
	Minimum detectable effect	0.068	0.066	0.059
S3-256	SQuAD 2.0	−0.027 [−0.104, +0.049]	−0.060 [−0.160, +0.060]	+0.060 [0.000, +0.140]
	COVID-QA	+0.005 [−0.050, +0.066]	−0.020 [−0.100, +0.040]	+0.060 [0.000, +0.140]
	CUAD	−0.008 [−0.027, +0.012]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
	Qasper	+0.045 [−0.017, +0.117]	+0.020 [0.000, +0.060]	+0.020 [0.000, +0.060]
	Pooled (stratified)	+0.004 [−0.024, +0.036]	−0.015 [−0.050, +0.020]	+0.035 [+0.015, +0.060]
	Minimum detectable effect	0.044	0.051	0.036

The P1 F1 MDE is 0.068 (computed from the standard deviation of all 200 queries; the MDE of the equal-weight estimand is 0.068, a post-hoc check). An MDE describes statistical sensitivity under a power calculation; it is not an observed bound that excludes larger effects. The confidence intervals are the basis for interpreting this experiment.

Output length also matters. Of the 400 CUAD outputs, 65 reached the generation cap, as did 2 of the 400 Qasper outputs (Table 15). A truncated output cannot match exactly, and the cap was reached more often in some cells than in others. In a post-hoc check, removing the 26 CUAD and 2 Qasper queries with any truncated output leaves every reading in Table 5 unchanged (P1 exact match −0.055, [−0.105, −0.010]). So does resampling documents instead of queries (−0.055, [−0.095, −0.020]), which matters because sampled queries share documents in SQuAD, COVID-QA and CUAD. The negative exact-match interaction comes from SQuAD (−0.160) and COVID-QA (−0.060); exact match is zero in every CUAD cell and rare for Qasper's free-form answers. For P1 on SQuAD at the matched budget (an exploratory observation), F1 is 0.58 with the longer passages and 0.44 with the shorter passages, at similar coverage (0.88 and 0.86). Passage continuity is a possible explanation, not an established mechanism.

7 Discussion

7.1 What changes when the protocol changes?

Count matching makes retrieval depth equal; budget matching makes the allowance in word occurrences equal. Neither holds every property of the context fixed. Longer passages may contain more evidence, but segmentation also affects ranking, redundancy, boundaries and continuity. The small overlap shifts and the direction of the unequal-length comparisons are consistent with text allowance contributing to protocol sensitivity. They do not identify it as the sole cause.

Budget matching also has costs. It can cut through the final passage, and equal word counts do not imply equal model-token counts, unique context, indexing cost or latency. These are reasons to specify the constraint, not reasons to avoid the comparison. Fixed-k evaluation answers a legitimate deployment question when the system retrieves a fixed number of passages. The problem arises when its answer is presented as an intrinsic property of a chunker.

The dense retriever begins with a different preference from the lexical grid. Under count matching, it supports the 128-word arm in 33.3% of universes and the 512-word arm in 27.1%, whereas the lexical grid supports them in 0.0% and 78.5%. Topic mixing within an embedding is one possible explanation, but our experiment does not measure that mechanism. What it establishes is configuration dependence.

7.2 Why retrieval evidence is not enough

The answer experiment adds a distinction. Increased gold-span coverage can coexist with unresolved answer-F1 effects and an exact-match interaction in the opposite direction. A reader may respond differently to passage continuity, distractors, answer length and normalization. The present check cannot separate those explanations, and its single-reader setting does not support a general claim that protocol effects stop at retrieval.

7.3 Reporting implications

A chunking comparison should specify nominal caps, realized length distributions, overlap handling, retriever, metric and operating point. For the retrieved context, it should report both passage count and word or token volume, including the treatment of repeated text and the final chunk. Showing count-matched and budget-matched results side by side makes their different constraints visible.

Statistical summaries need the same precision. Support shares require universe counts, grid weights and an uncertainty procedure. They should not be called flip rates without pairing universes and measuring transitions. Query-level and document-level resampling should be distinguished, and one decision rule should define both a point estimate and its interval. Finally, retrieval and answer outcomes should be reported separately: neither substitutes for the other.

8 Limitations and threats to validity

Measurement. Our span metrics reward any qualifying or best-covered evidence span and do not penalize irrelevant context. They need not predict precision, complete multi-hop evidence acquisition, factuality or answer quality. Word budgets count repeated occurrences, whereas coverage is measured on unique word positions. The metrics are computed on words, not on characters as the plans state; a post-hoc character-based rescoring of the BM25 selections changed no hypothesis verdict, but its effect on the dense-retrieval results was not quantified. We did not evaluate precision-sensitive measures, such as token-level precision or a graded ranking metric over evidence, so we cannot say

how much of the budget-matched advantage of small chunks reflects better ranking rather than more independently ranked units.

Implementation and scope. Sentence splitting is rule-based, paragraph boundaries depend on blank lines and coincide with the datasets' annotation units, and budget selection can truncate a chunk. Each query was ranked only to a fixed depth derived from corpus chunk lengths; in 2 stored selections (dense retrieval, Qasper, 2,048-word budget) that prefix held 1,880 and 1,789 words, short of the budget, and no other stored selection fell short. Setting both selections to the maximum possible scores changes no hypothesis verdict or estimate and moves interval bounds by at most 0.1 percentage points (a post-hoc bound; the dense ranking could not be re-executed). Dense rankings use the dot product of half-precision unit vectors, so near-ties can be reordered. In 12 of the 2,202 Qasper questions, an evidence paragraph occurs twice in its paper and only the first occurrence counts as gold. Each semantic chunker uses one setting. The clustering arm is mostly non-contiguous and close to its cap. Same-cap overlap contrasts differ in realized lengths and redundancy, so they are not exact mechanistic controls. The study covers four English collections, one lexical family and one dense encoder. Study 1b reuses Study 1's confirmatory queries. The answer check uses one quantized reader, a small query sample and an output cap that truncated some responses.

Inference. Universes are correlated settings, not independent replications. A universe's verdict depends on its sample size as well as on its effect, so support shares behave like vote counts: a dataset with more queries (SQuAD, 4,744, against 996 for CUAD) yields decisive verdicts for smaller effects. Mean differences per dataset are in Table 9. Joint resampling preserves their shared-query dependence but does not sample new datasets. We report query- and document-level inference; SQuAD has only 28 confirmatory documents, so its document-level intervals rest on few clusters. The per-universe intervals rely on approximate normality of mean differences, and the share intervals are approximate for the reasons given in Section 4. Support shares depend on the grid, its weights and the significance rule. H3 uses an approximate power calculation. The answer experiment is exploratory, not a multiplicity-controlled family of confirmatory tests.

Study history. H5b was added after the pilot, and one document-bootstrap sensitivity interval was corrected after the Study 1 run; both are listed in Appendix A. The inference procedure was corrected after all results were known; Table 7 keeps the plans' computation alongside. The stored analysis of the Study 1 pilot predates two logged changes (sensitivity intervals from 200 rather than 1,000 joint replicates, and query resampling inside the document-cluster interval). With those settings its 8 affected interval bounds reproduce exactly; the current code moves them by at most 1.85 percentage points and changes no estimate or verdict. The analysis plans were not deposited with an external registry, so the dates in their deviation logs are the project's own records. The post-hoc checks reported here (metric breakdown, gold-span containment, character-based metrics, the underfill bound and the answer-check sensitivity analyses) were chosen after the results were known.

9 Conclusion

Chunking verdicts depend on the retrieval-selection protocol. Matching words instead of chunk count reduces support for the larger fixed arm by 70.9 pp in the primary lexical comparison. In breakpoint comparisons, the shorter arm gains 55.6 pp. Both hold with documents as the sampling

unit. These are changes in support shares within the evaluated grid, not universal rankings of chunkers. The original dense-retrieval hypothesis is unresolved, because its verdict depends on the inference unit, while its disclosed shorter-arm amendment, formulated after the pilot, is supported at both units.

The answer-quality check does not detect a corresponding positive F1 interaction, although its intervals exclude interactions as large as its minimum detectable effects; for the primary comparison, exact match moves in the opposite direction. The practical lesson is to state what the comparison holds constant: report retrieval depth and text budget, and test answer quality rather than infer it from evidence coverage.

10 Reproducibility

The code, the analysis plans with their deviation logs, dataset manifests with SHA-256 digests of the source files, run configurations, per-query results and analysis outputs accompany this paper and are available at `https://github.com/swaroopchandjc/rag-research`. Every result value in the text, tables and figures is generated from the stored result files, and a separate script recomputes each reported value from its source; design constants (dataset names, grid sizes, seeds, thresholds) are written directly in the text.

Recomputing an analysis from stored outputs is not the same as re-running an experiment. From the stored per-query results, the planned analyses of the confirmatory runs and of the answer check recompute without difference, the stored pilot analysis recomputes except for the 8 interval bounds described in Section 8, and the reported analyses were also reproduced by an independent implementation of the verdict rule and the resampling. Of the experiments, only BM25 retrieval was re-executed from the public data; it reproduced every stored score and word count. Dense retrieval and answer generation need a GPU and were not re-executed; half-precision embeddings and greedy decoding can differ across hardware. The stored runs predate the run format that records digests of the code and inputs.

Use of generative AI tools

The author designed the study, wrote the core experimental pipeline (chunking, retrieval, evaluation, semantic chunking, dense retrieval, generation and run code), made the methodological and experimental decisions, ran the experiments, read the cited literature and interpreted the results. An AI assistant (Claude, Anthropic) was used under the author's direction to help draft the Study 1b analysis plan, run literature searches, debug and revise parts of the code, implement parts of the statistical-analysis, validation and test code, draft and revise manuscript text, and check the reported computations. The author reviewed all AI-produced code and text and takes full responsibility for the content of the paper.

References

[1] Samuel J. Bell, Onno Kampman, Jesse Dodge, and Neil D. Lawrence. Modeling the Machine Learning Multiverse. In *NeurIPS*, 2022. URL `https://proceedings.neurips.cc/paper_files/paper/2022/hash/750337e1301941f81ae31a90e0a1c181-Abstract.html`. arXiv:2206.05985.

[2] Yoav Benjamini and Yosef Hochberg. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1): 289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.

[3] Sinchana Ramakanth Bhat, Max Rudat, Jannis Spiekermann, and Nicolas Flores-Herr. Rethinking Chunk Size For Long-Document Retrieval: A Multi-Dataset Analysis. arXiv preprint arXiv:2505.21700, 2025. URL https://arxiv.org/abs/2505.21700.

[4] Jianlyu Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. M3-Embedding: Multi-Linguality, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation. In *Findings of ACL*, pages 2318–2335, 2024. doi: 10.18653/v1/2024.findings-acl.137. URL https://aclanthology.org/2024.findings-acl.137/. arXiv:2402.03216.

[5] Tong Chen, Hongwei Wang, Sihao Chen, Wenhao Yu, Kaixin Ma, Xinran Zhao, Hongming Zhang, and Dong Yu. Dense X Retrieval: What Retrieval Granularity Should We Use? In *EMNLP*, pages 15159–15177, 2024. doi: 10.18653/v1/2024.emnlp-main.845. URL https://aclanthology.org/2024.emnlp-main.845/. arXiv:2312.06648.

[6] Pradeep Dasigi, Kyle Lo, Iz Beltagy, Arman Cohan, Noah A. Smith, and Matt Gardner. A Dataset of Information-Seeking Questions and Answers Anchored in Research Papers. In *NAACL-HLT*, pages 4599–4610, 2021. doi: 10.18653/v1/2021.naacl-main.365. URL https://aclanthology.org/2021.naacl-main.365/. arXiv:2105.03011.

[7] Jesse Dodge, Suchin Gururangan, Dallas Card, Roy Schwartz, and Noah A. Smith. Show Your Work: Improved Reporting of Experimental Results. In *EMNLP-IJCNLP*, pages 2185–2194, 2019. doi: 10.18653/v1/D19-1224. URL https://aclanthology.org/D19-1224/. arXiv:1909.03004.

[8] André V. Duarte, João Marques, Miguel Graça, Miguel Freire, Lei Li, and Arlindo L. Oliveira. LumberChunker: Long-Form Narrative Document Segmentation. arXiv preprint arXiv:2406.17526, 2024. URL https://arxiv.org/abs/2406.17526.

[9] Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, Meng Wang, and Haofen Wang. Retrieval-Augmented Generation for Large Language Models: A Survey. arXiv preprint arXiv:2312.10997, 2023. URL https://arxiv.org/abs/2312.10997.

[10] Michael Günther, Isabelle Mohr, Daniel James Williams, Bo Wang, and Han Xiao. Late Chunking: Contextual Chunk Embeddings Using Long-Context Embedding Models. arXiv preprint arXiv:2409.04701, 2024. URL https://arxiv.org/abs/2409.04701.

[11] Dan Hendrycks, Collin Burns, Anya Chen, and Spencer Ball. CUAD: An Expert-Annotated NLP Dataset for Legal Contract Review. In *NeurIPS 2021 Datasets and Benchmarks Track*, 2021. URL https://arxiv.org/abs/2103.06268. arXiv:2103.06268.

[12] Antonio Jimeno Yepes, Yao You, Jan Milczek, Sebastian Laverde, and Renyu Li. Financial Report Chunking for Effective Retrieval Augmented Generation. arXiv preprint arXiv:2402.05131, 2024. URL https://arxiv.org/abs/2402.05131.

[13] Ehsan Kamalloo, Nouha Dziri, Charles L. A. Clarke, and Davood Rafiei. Evaluating Open-Domain Question Answering in the Era of Large Language Models. In *ACL*, pages 5591–5606, 2023. doi: 10.18653/v1/2023.acl-long.307. URL https://aclanthology.org/2023.acl-long.307/. arXiv:2305.06984.

[14] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *NeurIPS*, 2020. URL https://arxiv.org/abs/2005.11401. arXiv:2005.11401.

[15] Wensheng Lu, Keyu Chen, Ruizhi Qiao, and Xing Sun. HiChunk: Evaluating and Enhancing Retrieval-Augmented Generation with Hierarchical Chunking. arXiv preprint arXiv:2509.11552, 2025. URL https://arxiv.org/abs/2509.11552.

[16] Xing Han Lù. BM25S: Orders of magnitude faster lexical search via eager sparse scoring. arXiv preprint arXiv:2407.03618, 2024. URL https://arxiv.org/abs/2407.03618.

[17] Evan Miller. Adding Error Bars to Evals: A Statistical Approach to Language Model Evaluations. arXiv preprint arXiv:2411.00640, 2024. URL https://arxiv.org/abs/2411.00640.

[18] Timo Möller, Anthony Reina, Raghavan Jayakumar, and Malte Pietsch. COVID-QA: A Question Answering Dataset for COVID-19. In *Proceedings of the 1st Workshop on NLP for COVID-19 at ACL*, 2020. URL https://aclanthology.org/2020.nlpcovid19-acl.18/.

[19] Renyi Qu, Ruixuan Tu, and Forrest Sheng Bao. Is Semantic Chunking Worth the Computational Cost? In *Findings of NAACL*, pages 2155–2177, 2025. doi: 10.18653/v1/2025.findings-naacl.114. URL https://aclanthology.org/2025.findings-naacl.114. arXiv:2410.13070.

[20] Pranav Rajpurkar, Robin Jia, and Percy Liang. Know What You Don't Know: Unanswerable Questions for SQuAD. In *ACL 2018 (Volume 2: Short Papers)*, pages 784–789, 2018. doi: 10.18653/v1/P18-2124. URL https://aclanthology.org/P18-2124/. arXiv:1806.03822.

[21] David Rau, Hervé Déjean, Nadezhda Chirkova, Thibault Formal, Shuai Wang, Stéphane Clinchant, and Vassilina Nikoulina. BERGEN: A Benchmarking Library for Retrieval-Augmented Generation. In *Findings of EMNLP*, pages 7640–7663, 2024. doi: 10.18653/v1/2024.findings-emnlp.449. URL https://aclanthology.org/2024.findings-emnlp.449/. arXiv:2407.01102.

[22] Stephen Robertson and Hugo Zaragoza. The Probabilistic Relevance Framework: BM25 and Beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389, 2009. doi: 10.1561/1500000019. URL https://www.nowpublishers.com/article/Details/INR-019.

[23] Uri Simonsohn, J. Simmons, and Leif D. Nelson. Specification curve analysis. *Nature Human Behaviour*, 4(11):1208–1214, 2020. doi: 10.1038/s41562-020-0912-z.

[24] Mark D. Smucker, James Allan, and Ben Carterette. A Comparison of Statistical Significance Tests for Information Retrieval Evaluation. In *CIKM*, 2007. doi: 10.1145/1321440.1321528. URL https://ciir-publications.cs.umass.edu/getpdf.php?id=744.

[25] Sara Steegen, Francis Tuerlinckx, Andrew Gelman, and Wolf Vanpaemel. Increasing Transparency Through a Multiverse Analysis. *Perspectives on Psychological Science*, 11(5):702–712, 2016. doi: 10.1177/1745691616658637.

[26] Julián Urbano, Harlley Lima, and Alan Hanjalic. Statistical Significance Testing in Information Retrieval: An Empirical Analysis of Type I, Type II and Type III Errors. In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '19)*, 2019. doi: 10.1145/3331184.3331259. URL https://arxiv.org/abs/1905.11096. arXiv:1905.11096.

[27] An Yang et al. Qwen3 Technical Report. arXiv preprint arXiv:2505.09388, 2025. URL https://arxiv.org/abs/2505.09388.

[28] Jihao Zhao, Zhiyuan Ji, Zhaoxin Fan, Hanyu Wang, Simin Niu, Bo Tang, Feiyu Xiong, and Zhiyu Li. MoC: Mixtures of Text Chunking Learners for Retrieval-Augmented Generation System. arXiv preprint arXiv:2503.09600, 2025. URL https://arxiv.org/abs/2503.09600.

A Deviations from the analysis plans

Table 6 summarizes the changes logged in the two plans' deviation logs and the differences between the plans' text and the implementation. The timing column reports what the logs record. H5b adds a secondary hypothesis without replacing H5. The post-run Study 1 correction changes the document-cluster sensitivity interval, not its point estimate or verdict. The inference correction changes estimates by at most 2.1 percentage points and only H5's verdict (Table 7). Section-number errors in the plans' cross-references are listed in a separate erratum. Two exploratory items in the plans are not reported: the chunk-level reciprocal rank RR@10 of Study 1 (stored with the results) and an optional answer check for S5-256 in Study 1b, which was not run.

Table 6: Study-plan deviations (as logged) and differences between the plans' text and the implementation.

Study	Change	Timing as logged, and consequence
Study 1	Restrict main budgets to $B \geq \max(W_A, W_B)$; retain all budgets for sensitivity analysis	Before the pilot and confirmatory run; a smoke test showed that smaller budgets force truncation of the first chunk.
Study 1	Add document-cluster resampling	Before the runs; multiple queries are nested in a small number of SQuAD articles.
Study 1	Align the η^2 description with the analytic implementation	After the pilot and before the plan's final version; the estimand and thresholds are unchanged.
Study 1	Correct query-level resampling inside the document-cluster sensitivity interval	After the confirmatory run; the interval changes, while the point estimate and verdict do not.
Answer check	Planned: two hosted readers, 200 queries per dataset, $k \in \{3, 5\}$ and $B \in \{512, 1{,}024\}$, two samples per query. Executed: one local reader (Qwen3-4B, greedy), 50 queries per dataset, $k = 5$ and $B = 1{,}024$	Before any analyzed generation, following API-quota limits; the smaller sample lowers power, and the check remains exploratory.
Answer check	Two partial runs (21 hosted-model rows; 198 rows of a CPU run) were stopped and are not analyzed; the analyzed run is the first complete GPU run	Before the analyzed run, by the rule recorded in the log; CPU and GPU outputs are never pooled.
Study 1b	Specify implementation details	During implementation, before pilot output.
Study 1b	Add H5b and retain H5	After the exploratory pilot, before the confirmatory run; a new secondary estimand is disclosed.
Study 1b	Retain the clustering defaults despite near-cap chunks	After the pilot; the configuration is not tuned to improve the observed outcome.
Study 1b	The plan names both the confirmatory and the full corpus for the chunk statistics that orient the shorter arm	The implementation uses the full corpus; both give the same shorter arm in all 24 dataset-claim cases.
All	The metrics are computed on word intervals (gold character spans mapped to the words they overlap), whereas the plans define them over characters	The reported results are word-based; a post-hoc character-based rescoring of re-executed BM25 selections changes no verdict (Appendix D).
Study 1	The Benjamini–Hochberg family is claim, protocol and dataset; the plan says "within claim"	Pooling over datasets gives 70.2 pp instead of 67.6 pp.
Answer check	The reported MDE uses the pooled standard deviation of all queries	An MDE for the equal-weight estimand is reported alongside (post-hoc); values agree to about 0.001.
Study 1b	Budget-matched dense selections filled short of the budget: 2 (Section 8)	Post-hoc bound: no verdict changes.
Both	Inference: one Wald verdict rule for every estimate and bootstrap replicate, at the query or the document unit, instead of percentile verdicts for the estimate and normal-approximation verdicts with the query-level SD in the replicates	After all results were known; estimates move by at most 2.1 pp and only H5's verdict changes (Table 7).
Study 1	The stored pilot analysis predates two logged changes to the sensitivity intervals	8 interval bounds differ from a recomputation by at most 1.85 pp; estimates and verdicts are identical.

Table 7: Share hypotheses (pp, 95% CI, verdict) under the reported procedure (one Wald verdict rule for the estimate and every bootstrap replicate, query level), the same procedure with documents as the resampling and variance unit, and the computation specified in the analysis plans (percentile-bootstrap verdicts for the estimate, normal-approximation verdicts with the query-level SD in the replicates).

Hypothesis	Reported (query)	Document unit	Planned computation
H1 (P1 decline)	70.9 [64.8, 81.9] supported	68.9 [61.6, 80.9] supported	71.1 [64.8, 81.9] supported
H1b (S3-256 gain)	27.1 [15.4, 33.9] supported	26.4 [15.0, 34.4] supported	26.7 [15.4, 33.9] supported
H4 (breakpoint, shorter arm)	55.6 [50.8, 63.1] supported	59.0 [51.3, 65.0] supported	55.4 [50.8, 63.1] supported
H4b (clustering, shorter arm)	65.3 [56.5, 68.3] supported	65.2 [55.4, 67.1] supported	64.4 [56.5, 68.3] supported
H5 (dense P1 decline)	21.5 [16.7, 41.7] supported	19.4 [16.7, 39.6] inconclusive	19.4 [16.7, 41.7] inconclusive
H5b (dense P1, shorter arm)	58.3 [45.1, 64.6] supported	58.3 [47.9, 75.0] supported	58.3 [45.1, 64.6] supported

B Implementation details

Dense retrieval. The dense-retrieval code pins BGE-M3 at revision `5617a9f61b02`. The Study 1b run manifests record `sentence-transformers` 6.1.0, `transformers` 5.20.1 and `torch` 2.14.1 on a CUDA device in half precision. They do not record the model revision, so the run records alone cannot confirm that the stored runs used the pinned revision. Normalized embeddings are cached in half precision and ranked by their dot product without renormalization; the rounding changes vector norms by at most about 5×10^{-4}, so only near-ties can be reordered. Equal scores are ordered by the lower chunk index. Half-precision embeddings should not be assumed bit-identical across hardware.

Reader and prompt. The generation code pins the reader to the Qwen3-4B GGUF Q4_K_M file at revision `bc640142` and checks the file's SHA-256 before loading it; the generation records report `llama-cpp-python` 0.3.36 but not the revision or the file digest. The pinned model-file SHA-256 is:

`7485fe6f11af29433bc51cab58009521f205840f5b4ae3a32fa7f92e8534fdf5`

The ChatML prompt contains an empty thinking block. Decoding uses temperature zero, top-k one, seed 7 and the 256-token cap. The system instruction is:

> Answer the question using only the numbered passages. Reply with the shortest answer that is fully supported by the passages, copying the wording from them where possible. If the passages do not contain the answer, reply exactly: unanswerable

The user message lists numbered passages, followed by the question and an answer cue. Keeping this wording fixed prevents prompt changes from being mixed into the protocol contrast.

C Additional retrieval results

The tables below give corpus-level chunk lengths, per-dataset results, sensitivity analyses and power calculations. Corpus mean length should be distinguished from the amount of text actually selected for a query.

Table 8: Mean chunk length in words per chunking and dataset (full retrieval corpus), from the run manifests; they depend only on the chunking, not on any retrieval metric. These lengths define the "shorter arm" of every Study 1b claim. The fixed chunkings of Study 1b are the same chunks as in Study 1.

Chunking	SQuAD 2.0	COVID-QA	CUAD	Qasper
`fixed-64`	64	64	64	63
`fixed-128`	126	126	127	126
`fixed-256`	249	247	252	247
`fixed-512`	485	483	495	477
`fixed-64-o25`	64	64	64	64
`fixed-128-o25`	127	127	127	127
`fixed-256-o25`	252	252	253	251
`fixed-512-o25`	497	494	502	491
`sentence-64`	49	49	46	50
`sentence-128`	111	111	98	113
`sentence-256`	235	234	213	235
`sentence-512`	476	470	453	466
`paragraph-64`	46	45	41	42
`paragraph-128`	90	90	85	88
`paragraph-256`	187	179	174	199
`paragraph-512`	420	392	379	435
`semantic-bp`	293	287	320	278
`semantic-cl`	456	455	458	452
non-contiguous chunks (%)	92.5	94.1	92.0	94.7

Table 9: Study 1 per dataset: share of universes (%) supporting arm A or B, and the mean difference $\bar{\Delta}$ = metric(A) − metric(B) over universes. P1: A = fixed-128, B = fixed-512. S3-256: A = paragraph-256, B = fixed-256.

Claim	Dataset	A (count)	B (count)	$\bar{\Delta}$ (count)	A (budget)	B (budget)	$\bar{\Delta}$ (budget)
P1	SQuAD 2.0	0.0	100.0	−0.055	83.3	0.0	+0.053
	COVID-QA	0.0	89.1	−0.054	100.0	0.0	+0.109
	CUAD	0.0	25.0	−0.009	97.2	0.0	+0.040
	Qasper	0.0	100.0	−0.093	44.4	30.6	+0.002
S3-256	SQuAD 2.0	69.8	0.0	+0.014	100.0	0.0	+0.034
	COVID-QA	76.6	0.0	+0.028	100.0	0.0	+0.058
	CUAD	1.0	9.9	−0.003	9.4	2.1	+0.002
	Qasper	53.1	0.0	+0.019	99.5	0.0	+0.030

Table 10: Study 1, H1 under the planned sensitivity analyses: drop in the share of universes supporting fixed-512 from count- to budget-matched selection, in percentage points (threshold 20). (a) changes the verdict rule and has no joint-bootstrap CI.

Analysis	Estimate (pp)	95% CI
Main analysis	70.9	[64.8, 81.9]
(a) Benjamini–Hochberg FDR 5% within claim, protocol and dataset	67.6	—
(b) without SQuAD 2.0	61.2	[53.2, 74.8]
(c) default retriever only (Lucene, k_1=1.5, b=0.75, stemmed)	70.8	[64.6, 85.4]
(d) document resampling, cluster-robust intervals	68.9	[61.6, 80.9]
(e) all budgets, including $B <$ chunk size	73.9	[68.3, 84.8]

Table 11: Study 1, H2: main-effects decomposition of Δ_u for S3-256 under budget-matched selection.

Quantity	Value
η^2 of dataset	0.614
η^2 of budget	0.096
η^2 of retriever	0.044
η^2 of metric	0.032
dataset largest (share of bootstrap replicates)	100.0%

Table 12: Study 1, H3: paired queries needed to detect $|\delta| = 0.02$ ($\alpha = 0.05$ two-sided, power 0.8), $n = ((1.96 + 0.8416)\,\sigma_d/0.02)^2$, per budget-matched universe.

Claim	Median n	Universes with $n > 500$ (%)
S2-64	1,175	82.5
S2-128	1,162	84.8
S2-256	1,149	79.7
S2-512	1,161	78.0
S3-64	1,498	92.5
S3-128	1,412	90.6
S3-256	1,399	86.2
S3-512	1,248	84.7
All S2 and S3 universes	1,256	85.4

Table 13: Study 1b, H4 under the planned sensitivity analyses: share of S5 universes (%) supporting the shorter arm under count- and budget-matched selection, and the difference (pp; threshold 20). Study 1b uses only the default BM25 configuration, so sensitivity (c), the default-retriever restriction, does not apply.

Analysis	Count	Budget	Difference	95% CI
Main analysis (H4)	9.7	65.3	55.6	[50.8, 63.1]
(a) Benjamini–Hochberg FDR 5% within claim, protocol and dataset	—	—	54.3	—
(b) without SQuAD 2.0	7.4	55.6	48.1	[42.7, 58.3]
(d) document resampling, cluster-robust intervals	7.6	66.7	59.0	[51.3, 65.0]
(e) all budgets	9.7	63.3	53.6	[49.7, 60.9]
(f) BM25 only	0.0	61.1	61.1	[57.2, 72.5]
(f) dense only	19.4	69.4	50.0	[39.1, 58.1]

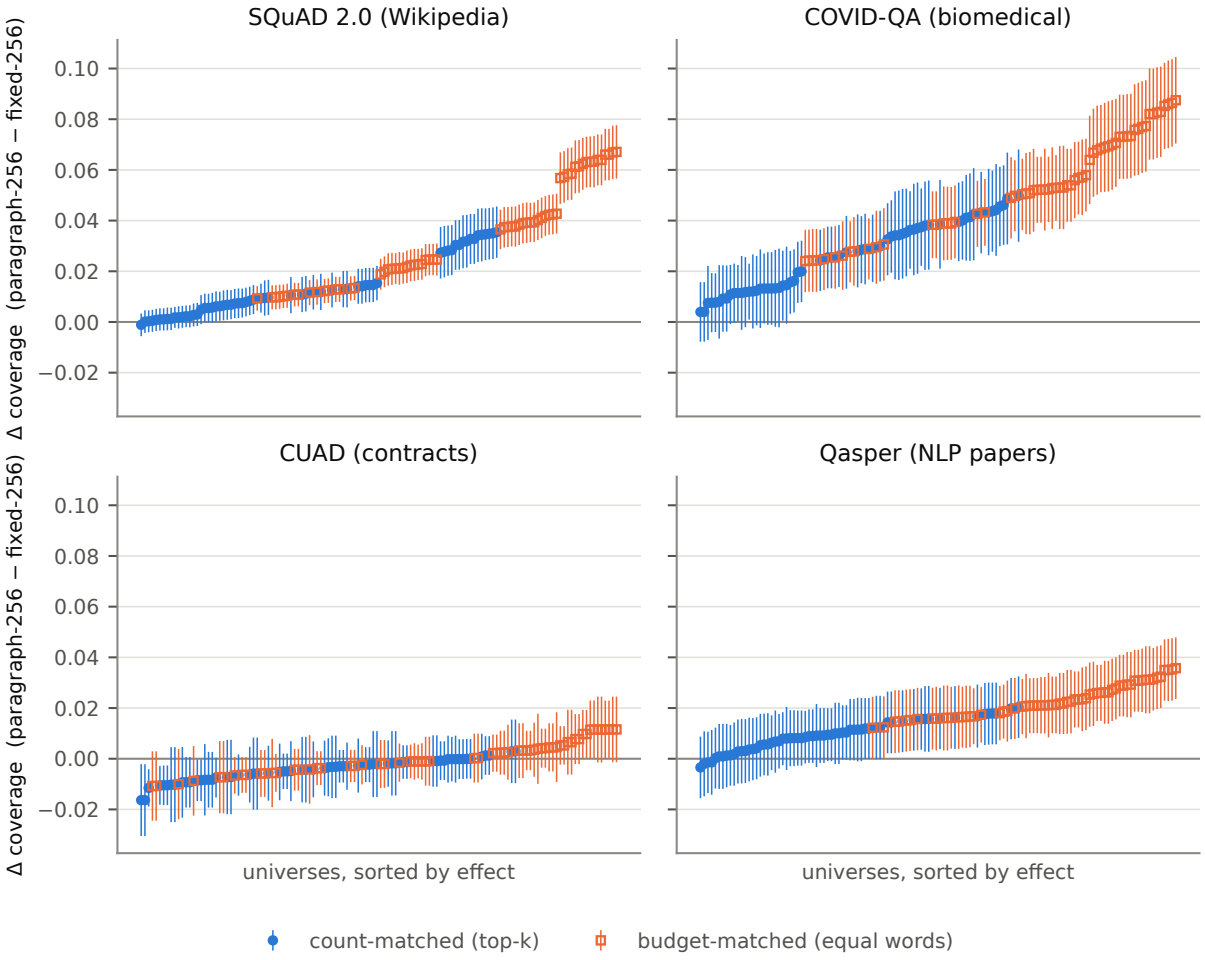


Figure 4: S3-256 coverage differences: paragraph packing minus fixed windows at a 256-word cap, with 95% Wald intervals (query level). Universes are sorted by effect within each dataset. The near-zero CUAD effects show that the paragraph-packing result is not uniform across domains.

D Post-hoc sensitivity: character-based metrics

This analysis is not part of the analysis plans. Rankings and selected chunks were not stored, so BM25 retrieval was re-executed with the stored configuration. Before attributing any difference to the metric, we checked that the re-executed word scores and word counts equal the stored arrays: all 85,275,648 stored Study 1 values and 1,665,540 stored Study 1b BM25 values match exactly. Matching scores and word counts make it very likely, but do not prove, that the same chunks were selected, because the stored runs contain no chunk identifiers to compare. The re-executed selections were then scored with literal characters (whitespace between separately retrieved chunks is covered by neither chunk) and with non-whitespace characters, and analyzed with the unchanged planned analysis code and thresholds. Table 14 reports the result. No hypothesis verdict changes. The literal-character definition changes 965 of 24,960 Study 1 universe verdicts, mostly for strict containment, and none reverses sign. Removing float16 storage changes none. The dense-retrieval arm of Study 1b was not re-executed, so the effect of the definition on H4, H4b, H5 and H5b as pooled over both retrievers is unquantified.

Table 14: Post-hoc sensitivity (not in the analysis plans): hypothesis estimates (pp, 95% CI) when the re-executed BM25 selections, which reproduce every stored score and word count, are scored on words (as reported), on words without float16 storage, on characters, and on non-whitespace characters. Study 1 columns: H1, H1b, H2 dataset η^2, H3 share of universes needing $n > 500$, and universe verdicts that change relative to the stored scores (of 24,960). Study 1b rows give the H4 and H4b forms restricted to BM25 universes (the dense arm was not re-executed). All rows use the computation specified in the analysis plans, so they are compared with the planned values.

Scoring	H1 / H4 (BM25)	H1b / H4b (BM25)	η^2 dataset	H3 $n > 500$ (%)	Changed verdicts
Stored (words, float16)	71.1 [64.8, 81.9]	26.7 [15.4, 33.9]	0.614	85.4	—
Words, float64	71.1 [64.8, 81.9]	26.7 [15.4, 33.9]	0.614	85.4	0
Characters	70.4 [63.9, 80.3]	24.2 [13.5, 30.7]	0.590	85.5	965
Non-space characters	71.1 [64.9, 81.9]	26.2 [15.0, 33.2]	0.620	85.4	197
Study 1b BM25: Stored (words, float16)	61.1 [57.2, 72.5]	63.9 [55.3, 71.1]			
Study 1b BM25: Words, float64	61.1 [57.2, 72.5]	63.9 [55.3, 71.1]			
Study 1b BM25: Characters	60.2 [54.9, 70.4]	60.2 [52.8, 67.8]			
Study 1b BM25: Non-space characters	62.0 [57.2, 72.5]	63.9 [55.3, 71.1]			

E Generation diagnostics

Table 15 summarizes completion, truncation and answer behavior over the 400 generations per dataset (50 queries, four chunkings, two operating points). The correlation is descriptive, not causal. A lenient hit means that at least half of one gold span was retrieved; it does not mean that all required evidence was present.

Table 15: Answer-quality check, planned diagnostics per dataset (400 generations each): completed jobs, answers truncated at the 256-token cap, share of "unanswerable" replies, and the link between retrieval and answers (Pearson correlation of coverage and F1; mean F1 when the gold evidence was / was not at least half retrieved).

Dataset	Jobs ok	Truncated	Unanswerable (%)	r(coverage, F1)	F1 | hit	F1 | miss
SQuAD 2.0	400/400	0	4.2	0.36	0.56	0.06
COVID-QA	400/400	0	5.8	0.33	0.40	0.15
CUAD	400/400	65	15.0	0.66	0.70	0.14
Qasper	400/400	2	15.8	0.53	0.37	0.08